

15-322 / 15-622 | FALL 2026

INTRODUCTION TO COMPUTER MUSIC

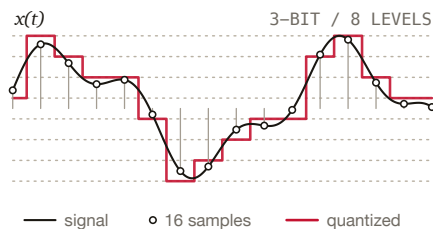
*Turn code into sound.*Sampling, synthesis, Fourier analysis, effects, and generative music,
from first principles to real-time systems.**TUE + THU**
11:00 AM – 12:20 PM**CIC 1203****9 UNITS · 15-322**
12 UNITS · 15-622

WHAT YOU'LL LEARN

01

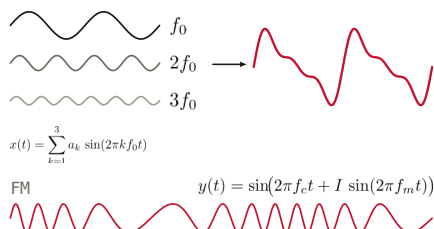
DIGITAL AUDIO

SAMPLING · QUANTIZATION · ALIASING

**02**

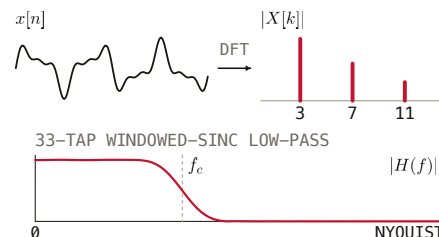
SYNTHESIS

ADDITIVE · FM · PHYSICAL MODELING

**03**

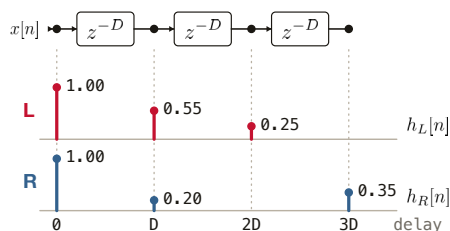
FOURIER & DSP

DFT / FFT · FILTERS · CONVOLUTION · STFT

**04**

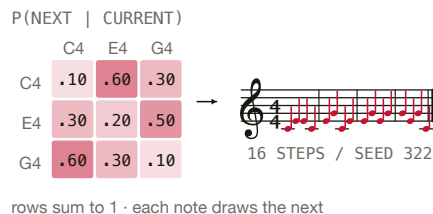
EFFECTS & SPATIALIZATION

DELAY LINES · MODULATION · PHASE VOCODING

**05**

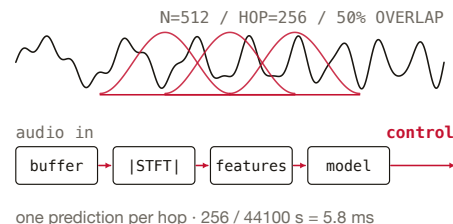
ALGORITHMIC COMPOSITION

MARKOV CHAINS · LPC · MIDI

**06**

REAL-TIME & MUSIC AI

STREAMING · MIR · GENERATIVE MODELS

**OPEN-ENDED PROJECTS**
EXPLORE YOUR OWN INTERESTS**PREREQUISITE**
15-122 OR 15-112**MUSIC EXPERIENCE**
NOT REQUIRED